

Project 4: Dashboard

STAT 133, Spring 2026

About

The purpose of this assignment is to give you the opportunity to create a **Quarto Dashboard** and also to put in practice the tidyverse tools that we've covered so far.

General Instructions

- Use a Quarto Document (.qmd file) for your Project 4. Please refer to the template files provided in lab 8 and in bCourses folder project4.
- Save this file as `project4.qmd`.
- You will have to Render your `qmd` file as a `dashboard` document.
- You will also have to publish your dashboard to **Netlify**.
- As usual, assume that your published dashboard will be seen by a potential employer, boss, client/customer, admissions committee, or something like that. In other words, make sure that the content of your project is “understandable”: well organized, self descriptive, easy to follow, etc.
- To receive full credit, please submit the following to bCourses (proj4 assignment):
 - qmd file
 - HTML file
 - (URL) link to your **Netlify** published document

1 Data

The data for this project is somewhat related to the data of Project-2. Specifically, the data involves crashes in California involving bicyclists between 2014 and 2023. All the files are available in bCourses.

<https://bcourses.berkeley.edu/courses/1551809/files/folder/projects/project4>

- CSV File 1: `bicyclist_alpha_2014_2023.csv`
 - This file contains information about the collision date and time.
- CSV File 2: `bicyclist_beta_2014_2023.csv`
 - This file contains information about the type of collision, the violation category, party at fault, etc.
- PDF Data Documentation: `Codebook-Bicyclists-Crash-Data-2014-2023.pdf`

Importing CSV files

We recommend importing the CSV files in R using the `read_csv()` function from *tidyverse* package "readr".

1.1 Data Preparation on `bicyclist_alpha_2014_2023.csv`

Before you begin your exploratory-and-descriptive analysis, you will have to carry out some *data preparation* steps regarding `bicyclist_alpha_2014_2023.csv`. To be more precise, after importing this data in R you will have to add (i.e. `mutate()`) the following time-date-related columns:

- **YEAR**: use the `year()` function from *tidyverse* "lubridate" to extract the year from column `COLLISION_DATE`.
- **MONTH**: use the `month()` function from *tidyverse* "lubridate" to extract the month from column `COLLISION_DATE`.
- **DAY_OF_WEEK**: use the `wday(..., label = TRUE)` function from *tidyverse* "lubridate" to extract the day of the week from column `COLLISION_DATE`. And the mutate it into a factor specifying `levels` in such a way that the values are: "Mon" "Tue" "Wed" "Thu" "Fri" "Sat" "Sun".
- **HOURL**: use the function `case_when()` to mutate a column `HOURL` by categorizing the `COLLISION_TIME` into one-hour values; see sample code below:

```
HOURL = case_when(  
  COLLISION_TIME >= 2300 ~ 23,  
  COLLISION_TIME >= 2200 ~ 22,  
  COLLISION_TIME >= 2100 ~ 21,  
  ...  
  COLLISION_TIME >= 200 ~ 2,  
  COLLISION_TIME >= 100 ~ 1,  
  COLLISION_TIME >= 0 ~ 0)
```

1.2 Joining alpha and beta tables

Merge (i.e. *join*) the **alpha** tibble of collision date-time information with the **beta** tibble from the `bicyclists_beta_2024_2023.csv` file. Use the `CASE_ID` column as the *ID* column, that is, the column that both **alpha** and **beta** files have in common.

2 Suggested Dashboard Topics

Dashboard Topics

The goal is to create a dashboard that addresses two of the topics listed below (1 to 6).

- Keep in mind that each topic is fairly open-ended. This is intentionally: to give you plenty of room for exploration and creativity.
- Likewise, each research question should consist of two perspectives:
 - a) analysis on **ALL** collisions (of both non-fatal and fatal outcomes)
 - b) comparative analysis between non-fatal -vs- fatal outcomes

Topics

- 1) **Long-term Trends:** Over the 10-year period (2014-2023), what has been the trend of bicyclist collisions?
 - Overall trend (all collisions).
 - Trend of killed bicyclists.
 - Has the ratio of “killed” to “injured” bicyclists decreased?
- 2) **Seasonality Trends:** How do collision rates fluctuate across “seasons”? Notice that “seasons” is left to your own interpretation. For example: it could be annual seasons (spring, summer, fall, winter), or monthly, or vacations -vs- non-vacation periods, etc.
 - Overall trend (all collisions).
 - Trend of killed bicyclists.
 - Has the ratio of “killed” to “injured” bicyclists decreased?
- 3) **The Weekend -vs- Weekday Collisions:** Is there a difference in the number of bicyclist collisions on weekends versus weekdays?
 - All collisions.
 - Killed bicyclists -vs- Injured bicyclists?
- 4) **The “Rush Hour” Effect:** Using time, what is the distribution of collisions throughout the day? Is there a rush hour effect? Are there certain hours where the number of bicyclist collisions tend to increase?
 - All collisions.
 - Killed bicyclists -vs- Injured bicyclists?
- 5) **Ranking of Collision Types:** What are the most common types of collision? What are the least common ones?
 - Ranking types of collision among all collisions.
 - Ranking types of collision among Injured bicyclists.
 - Ranking types of collision among Killed bicyclists.
- 6) **The Party at Fault Association:** What is the association between the primary cause (i.e. Violation Category) and which party is found at fault?
 - Association considering all collisions.
 - Association considering non-fatal outcomes.
 - Association considering fatal outcomes.

💡 Specs

Make sure your dashboard meets the following specifications:

- This Dashboard is NOT a *Shiny* dashboard (i.e. don't use interactive widgets since this will be part of project 5).
- Your dashboard should contain **at least 2 main tabs**.
- You can include **sub-tabs** in any of your main tabs.
- You can choose **any kind of layout**.
- Include 2 or more **Value Boxes** in at least one of your tabs or sub-tabs.
- Your dashboard should include at least one graphic containing **facets**.
- Your dashboard should include at least one **output table**.
- All **graphics have to be interactive**: your ggplot graphics have to be passed to either the "plotly" function `ggplotly()` or using "ggiraph" layers and `girafe()`.
- Likewise, graphics have to be presentation-quality plots.

Demo Dashboard

You can find a dashboard demo that uses Project-2 data `crashes_2023.csv` (see bCourses Files/projectst/project-4/dashboard-demo-2023)

<https://bcourses.berkeley.edu/courses/1551809/files/folder/projects/project4/dashboard-demo-2023>

! Important

You can use the provided demo file as a conceptual guide to help you get started, but it should serve only as inspiration. Your final submission must be an original implementation that demonstrates your own logic and problem-solving. Simply adapting or refactoring the demo code to fit the assignment requirements is not permitted; your work must go significantly beyond the demo's scope to earn full credit.

Quarto Dashboards ocumentation

Here are some resources about Quarto dashboards that you may find useful:

- Quarto dashboards documentation: <https://quarto.org/docs/dashboards/>
- Dashboard Layout: <https://quarto.org/docs/dashboards/layout.html>
- Dashboard Data Display: <https://quarto.org/docs/dashboards/data-display.html>
- Example: New Zealand Earthquakes (by Charlotte Wickham)
 - <https://pub.demo.posit.team/public/nz-quakes/quakes.html>
 - <https://github.com/cwickham/quakes/blob/main/quakes.qmd>
- Example: Norfolk dashboard (by Garrick Aden-Buie)
 - <https://grrrck.quarto.pub/mynorfolk-dash/>
 - <https://github.com/gadenbuie/mynorfolk-dash/blob/main/index.qmd>